Tuning-Fork Calibration: Fitting a Pure Sine to Sampled Data

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Abstract

A tuning fork is struck and the displacement is sampled at fixed intervals. We fit the sampled points to the pure sine model and use the fitted frequency as the calibration value.

1 The model

The displacement follows

$$y = a \sin(bx), \tag{1}$$

with amplitude a and angular frequency b; the independent variable x is time in seconds.

2 Sampled points

The calibration data used for the fit are:

x (s)	y (mm)
0.0	0.00
0.5	1.42
1.0	2.00
1.5	1.41
2.0	0.03

3 Calibration remark

The fit residual against the sampled data determines whether a second harmonic is present; for the fork in question the pure sine suffices.